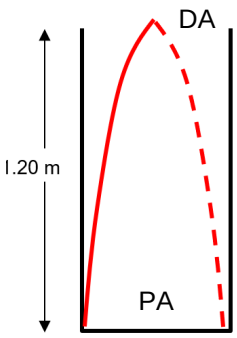
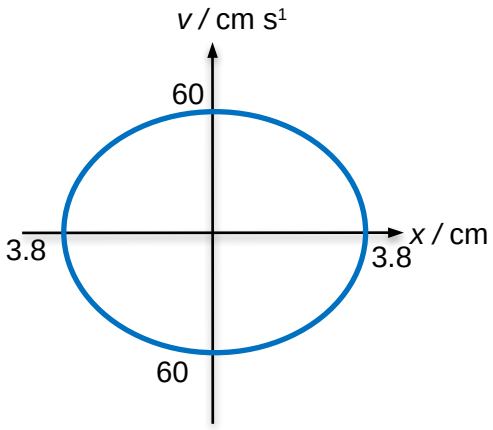
Annotations used in marking
BOD - Benefit of doubt
ECF - Error carried forward
POT - Powers of ten error
TE - Transfer error
CE - Calculation error
XP - Wrong physics
ENG - Generally bad english, phrasing and expression
PP - Poor presentation of answers

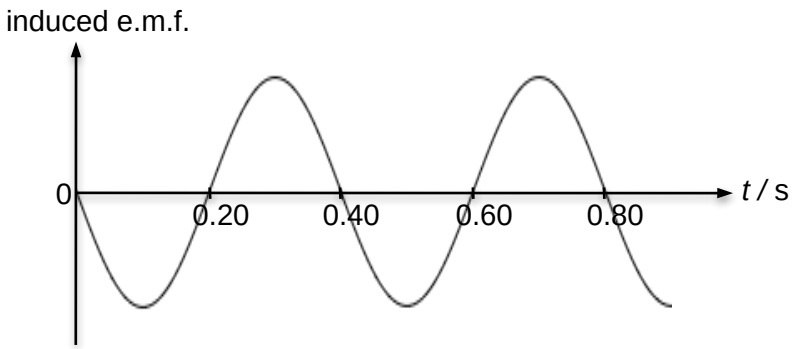
Qn	Suggested Answer
1	
(a)	Base SI Units of P = kg m^{-1} s^{-2}
	Units of $\eta = \dfrac{(\text{units of } P)(\text{units of } r)^4}{(\text{units of } l)(\text{units of } \frac{V}{t})}$ $= \dfrac{(\text{kg m}^{-1}\ \text{s}^{-2})(\text{m})^4}{(\text{m})(\text{m}^3\ \text{s}^{-1})}$
	$=$ kg m^{-1} s^{-1}
(b)(i)	$\pm\dfrac{\Delta\eta}{\eta} x100\% = \pm(\dfrac{\Delta P}{P} + 4(\dfrac{\Delta r}{r}) + \dfrac{\Delta l}{l} + \dfrac{\Delta(\frac{V}{t})}{(\frac{V}{t})})x100\%$
	$= \pm(5\% + 4(0.5\%) + 2\% + 3\%)$
	$= \pm 12\%$
(b)(ii)	$\pm\Delta\eta = 0.12 \times 1.137 \times 10^{-3} = 0.136 \times 10^{-3}$ kg m^{-1} s^{-1} Ans: $(1.1 \pm 0.1) \times 10^{-3}$ kg m^{-1} s^{-1}
2	
(a)	Electric field strength at a point is defined as the electric force per unit charge
	on a positive charge placed at that point.
(b)(i)	$E = \dfrac{1}{4\pi\varepsilon_o}\dfrac{Q}{r^2}$ $= \dfrac{1}{4\pi\varepsilon_o}\dfrac{(1.6\times10^{-19})}{(2.8\times10^{-10})^2}$
	$E = 1.8\times10^{10}$ N C^{-1}
(b)(ii)	$F = qE$ $= (1.6\times10^{-19})(1.835\times10^{10})$ $= 2.9\times10^{-9}$ N
(c)(i)	Nearest atoms are negatively charged and will exert an attractive force on the ion.
	Next nearest atoms are positively charged and will exert an repulsive force on the ion.
	However, due to the symmetry, for every force there is another force in the opposite

	direction of the same magnitude. Hence the resultant force is zero.
(c)(ii)	The central ion will vibrate about its equilibrium position.
(c)(iii)	Lowest electrical potential energy.
(d)	As the equilibrium distance for sodium chloride is shorter, the electrical potential energy is more negative.
	Since the change in potential energy is higher, a higher temperature is needed to separate the ions when melting.
3	
(a)(i)1.	m is the mass of a single gas molecule.
(a)(i)2.	$\langle c^2 \rangle$ is the mean squared speed of the gas molecules.
(a)(i)3.	$\frac{1}{2}m\langle c^2 \rangle$ is the mean kinetic energy of a gas molecule.
(a)(ii)	$U = \frac{3}{2}N_A kT$ $= \frac{3}{2}(6.02 \times 10^{23})(1.38 \times 10^{-23})(273.15 + 27.0)$ OR $U = \frac{3}{2}nRT$ $= \frac{3}{2}(1)(8.31)(273.15 + 27.0)$
	$= 3740$ J
(b)	At the same temperature, average kinetic energy of the molecules are the same. $\frac{1}{2}m_N c_{rms,N}^2 = \frac{1}{2}m_O c_{rms,O}^2$ $\frac{c_{rms,N}}{c_{rms,O}} = \sqrt{\frac{m_O}{m_N}}$ $= \sqrt{\frac{5.31 \times 10^{-26}}{4.65 \times 10^{-26}}}$
	$= 1.07$
(c)(i)	Work done by the gas = $p\,\Delta V$ $= (8.00 \times 10^6)(7.10 - 3.00) \times 10^{-5}$
	$= 328$ J
(c)(ii)	$Q = mc\Delta\theta$ $= (1.27 \times 10^{-3})(1.01 \times 10^3)(900)$
	$= 1.15443 \times 10^3$ J $= 1.15 \times 10^3$ J
(c)(iii)	$\Delta U = W + Q$ $= -328 + 1154$ $= 826$ J
4	
(a)	Coherent sources / constant phase relationship / same frequency
	Same amplitude / amplitude very close to each other
(b)(i)	$\frac{\lambda}{a} = \frac{\Delta x}{D}$ (Δx is fringe separation)

	$a = \frac{633 \times 10^{-9} \times 2.50}{(1.85 \times 10^{-2})/3}$
	$= 2.566 \times 10^{-4}$ m = 0.257 mm (3 sf)
(b)(ii)	*a* increases, but λ and *D* remains the same.
	$\Delta x = \frac{\lambda D}{a}$ As ( ), Δx decreases, hence the distance between the fringes decreases.
(b)(iii)	Hence, the fluctuation in *n* will result in random variations / changes / fuctuations in the wavelengths from each slit. *n* increases, λ decreases; *n* decreases, λ increases.
	The interference pattern will shift laterally with time / describe how positions of maxima will change / describe how distances between maxima changes OR Contrast of the interference pattern will decrease.
(c)(i)	
1.	PA and DA labelled correctly
2.	Pressure variation graph correct. (Need not touch the ends of pipe)
(c)(ii)	$L = \frac{\lambda}{4}$ $f_o = \frac{v}{4L}$ $= \frac{348}{4 \times 1.20}$ $= 72.5$ Hz
(c)(iii)	$L = \frac{3\lambda}{4}$ $f = \frac{3v}{4L} = 3f_o$ $= \frac{3 \times 348}{4 \times 1.20}$
	$= 217.5$ Hz $= 218$ Hz

5	
(a)	$E = pc$
(b)(i)	$\Phi = 2.30 \times 1.60 \times 10^{-19}$ $= 3.68 \times 10^{-19}$ J
	$f_0 = \frac{\Phi}{h}$ $= \frac{3.68 \times 10^{-19}}{6.63 \times 10^{-34}}$ $= 5.55 \times 10^{14}$ Hz
(ii)	$E_n = \frac{h^2}{8mL^2} n^2$ $= \frac{(6.63 \times 10^{-34})^2}{8(9.11 \times 10^{-31})(3.0 \times 10^{-9})^2} n^2$
	$\Delta E = \frac{(6.63 \times 10^{-34})^2}{8(9.11 \times 10^{-31})(3.0 \times 10^{-9})^2} (8^2 - 2^2)$ $= 4.021 \times 10^{-19}$ J
	$\Delta E = \frac{4.021 \times 10^{-19}}{1.60 \times 10^{-19}}$ $= 2.513$ eV $= 2.51$ eV
(iii)	$E_k = 2.51 - 2.30$ $= 0.21$ eV
	$E_k = \frac{p^2}{2m}$ $p = \sqrt{2mE_k}$ $= \sqrt{2(9.11 \times 10^{-31})(0.21 \times 1.60 \times 10^{-19})}$ $= 2.4743 \times 10^{-25}$ kg m s^{-1}
	$\lambda = \frac{h}{p}$ $= \frac{6.63 \times 10^{-34}}{2.4743 \times 10^{-25}}$ $= 2.68 \times 10^{-9}$ m $= 2.7 \times 10^{-9}$ m
(iv)	The electrons incident on the double slit have a range of speeds / kinetic energies.
	Hence, electrons have a range of de Broglie wavelengths, producing fringes with different spacings.
	The fringes overlap causing the interference pattern becomes less distinct / blurred.

6	
(a)	The resultant force / acceleration is always directed toward the equilibrium position/fixed point
	The resultant force / acceleration is directly proportional to the displacement / distance from the equilibrium positon/fixed point.
(b)(i)1.	$\omega = \frac{2\pi}{T}$ $= \frac{2\pi}{0.40}$
	$= 15.7 \text{ rad s}^{-1}$
(b)(i)2.	When displacement is zero, acceleration = 0
	When displacement is maximum, $a = \omega^2 x_0$ $= 15.708^2 \times 0.038$
	$= 9.3761$ $= 9.38 \text{ m s}^{-1}$
(b)(ii)	
	Shape of graph
	Labelling of *x*-axis and *v*-axis with values
(b)(iii)	$v = \omega\sqrt{x_0^2 - x^2}$ $\frac{1}{2}\omega x_0 = \omega\sqrt{x_0^2 - x^2}$
	$0.25x_0^2 = x_0^2 - x^2$ $x = \left(\sqrt{0.75}\right)(3.8)$
	$= 3.29 \text{ cm}$
	Alternatively, $x = -x_0 \cos \omega t$ $v = v_0 \sin \omega t$ $\frac{1}{2} v_0 = v_0 \sin \omega t$
	$t = 0.0333$ s

	$x = 3.8\cos(15.7 \times 0.0333)$ $= 3.29$ cm
(c)(i)	
	Shape of graph: sine/-sine
	Correct period
(c)(ii)	At the top or bottom of the oscillation, the magnet is momentarily at rest and hence, there is no change in magnetic flux linkage and there is no e.m.f. produced.
	At the equilibrium position, the velocity of the magnet is maximum and hence rate of change of flux linkage (or rate of cutting of flux) is a maximum. According to Faraday's Law, the induced e.m.f. is maximum.
	According to Lenz's Law, when the magnetic flux linkage increases, the induced e.m.f. will be such that the induced current if it flows will be such as to oppose the increase in magnetic flux linkage. Hence the induced e.m.f. have opposite signs when the magnet is moving down and when it is moving up.
(c)(iii)	(When the resistor is connected across XY, the circuit is closed.) There will now be an induced current flowing in the coil,
	producing a magnetic force that opposes the oscillation / energy is dissipated as thermal energy in the coil.
	Hence the amplitude of the oscillations decreases continuously over time.
7	
(a)(i)	$z = 92 - 42 = 50$
	$x = 235 + 1 - 99 - 134 = 3$
(a)(ii)	energy released $= (8.61 \times 99) + (8.45 \times 134) - (7.59 \times 235)$
	$= 201.04$ $= 201$ MeV (3 sf)
(a)(iii)	The reaction produces more neutrons which can react with Uranium nuclides to produce even more neutrons.
(b)(i)1	It is impossible to predict when a particular nucleus in a sample is going to decay. OR Each nucleus in a sample has the same chance of decaying per unit time.
(b)(i)2	Not affected by external factors such as presence of other nuclei, temperature and pressure.
(b)(ii)	${}^{99}_{42}\text{Mo} \rightarrow {}^{99m}_{43}\text{Tc} + {}^{0}_{-1}\text{e} + \bar{\nu} + \text{energy}$
(b)(iii)	mass defect $= (98.90771u - 98.90641u - 0.00055u)$ $= 7.5 \times 10^{-4} u$

	energy released $=(7.5\times10^{-4})(1.66\times10^{-27})(3.00\times10^{8})^{2}$ $=1.1205\times10^{-13}$ J
	$=\dfrac{1.1205\times10^{-13}}{10^{6}\times1.60\times10^{-19}}$ MeV $=0.70031$ MeV $=0.700$ MeV (3 sf)
(c)(i)	Probability per unit time that a nucleus will undergo decay.
(c)(ii)	$\lambda=\dfrac{\ln 2}{t_{\frac{1}{2}}}=\dfrac{\ln 2}{66\times60\times60}=2.91729\times10^{-6}$ s^{-1}
	$N=\dfrac{A}{\lambda}=\dfrac{12.0\times10^{9}}{2.91729\times10^{-6}}=4.11341\times10^{15}$
	mass $=4.11341\times10^{15}\times98.90771\times1.66\times10^{-27}$ $=6.75367\times10^{-10}$ $=6.75\times10^{-10}$ kg (3 sf)
(c)(iii)	$m=m_{0}e^{-\frac{\ln 2}{t_{\frac{1}{2}}}t}$ $=6.75\times10^{-10}e^{-\frac{\ln 2}{\left(\frac{66}{24}\right)}30}$
	$=3.51027\times10^{-13}$ $=3.51\times10^{-13}$ kg (3 sf)
(d)(i)	Short half-life of 6 hours allows sufficient time to perform scan OR It decays quickly enough to reduce radiation dose. OR γ-radiation is weakly ionising, which minimises tissue damage. OR γ-radiation is highly penetrating, which allows it to be easily detected outside the patient.
(d)(ii)	Decay constant is extremely small, activity inside the patient is negligible.
(d)(iii)	Industrial tracking takes place over long distances / long periods of time. A short half-life would decay to undetectable levels before process is complete.